Iterative Sample Generation and Balance Approach for Improving Hyperspectral Remote Sensing Imagery Classification With Deep Learning Network

Zhiyong Lv, *Senior Member, IEEE*, Pengfei Zhang, Linfu Xie, Jón Atli Benediktsson, *Fellow, IEEE*, and Tao Lei, *Senior Member, IEEE*

***Abstract*— Sample augmentation is effective for improving the supervised performance of land-cover classification with hyperspectral remote sensed image (HRSI) when the training samples are limited. However, numerous existing methods have neglected, considering the interclass-imbalance problem in the process of sample augmentation. In this work, new sample generation and sample balance strategies were promoted and simultaneously combined into an iteration for balancing and improving classification performance with HRSI. First, a sample augmentation with superpixel's constraint (SASC) is designed to augment the initial training samples set to avoid the overfitting of a sample generation neural network. Second, sample generation based on generative adversarial network (SGGAN) was proposed to generate samples for each class. Then, the proposed SASC, SGGAN, and a pattern recognition neural network named 3 dimensions-convolutional neural network (3-D-CNN) are combined into an iterative classification process called iterative sample generation and balance (ISGB) for balancing the user's accuracy for each class and optimizing the classification performance. Experiments on four widely used HRSIs are performed. The results when compared with eight state-of-the-art methods based on few-shot learning and generative adversarial network (GAN) efficiently demonstrate the feasibility and superiorities of the proposed approach for improving land-cover classification performance when the initial samples are limited. Moreover, the comparisons of the standard deviation of the user's accuracies (SDUA) demonstrated the balancing ability of the proposed approach. The code of the proposed approach is available at https://github.com/ImgSciGroup/ISGBA.**

***Index Terms*— Few-shot learning, hyperspectral remote sensing imagery, land-cover classification, sample augmentation.**

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I. Introduction

LAND-COVER classification with hyperspectral remote sensed images (HRSIs) aims at assigning a semantic label to each pixel in an entire image scene [1]. This mechanism plays an important role in land-use analysis [2], cropland mapping [3], and environmental health evaluation [4]. In the past decades, various land-cover classification approaches for HRSI have been promoted and successfully applied in practical applications [5], [6], such as supervised-based [7], semisupervised-based [8], and unsupervised-based [1] HRSI classification. Among these existing methods, without regarding the traditional methods (nondeep learning) and deep-learning-based neural networks for classification with HRSIs, the classification performance depends on the training samples for supervised classifiers and neural networks. However, labeling samples for an HRSI requires professional expertise and is time-consuming and labor-intensive. The challenge of labeling samples for the task of classification with HRSIs is briefly summarized below.

1) Manual labeling of numerous samples for an HRSI is time-consuming and labor-intensive [9], [10]. In practical applications, the labeling of samples is commonly referred to as "visual interpretation," necessitating the expertise of practitioners with a professional background. Besides visual interpretation, conducting field surveys with GPS equipment can also be used for sample labeling. However, errors in equipment are inevitable when labeling samples by comparing the position in an

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image and the corresponding position from a GPS [11]. In certain application scenarios, such as wetlands, rivers, and the tops of buildings, conducting ground field surveys poses significant challenges [12]. Therefore, labeling many samples for HRSI classification is usually time-consuming and labor-intensive.

2) Manual labeling of samples of an HRSI is difficult, making it challenging to mitigate the negative influence of interclass imbalance on the classification performance [13], [14]. Numerous existing studies have focused on modifying classifiers, optimizing neural networks, and enhancing the quality of samples [15]. However, the phenomenon of interclass imbalance may cause a bias in user accuracy when achieving the classification task with an equal quantity of samples for each class [16]. Therefore, in the practical reequipment of classification task with HRSIs, addressing the influence of interclass imbalance when labeling samples becomes crucial.
3) The high-dimensional characteristics of HRSI may result in mislabeling for certain samples when labeling samples manually [17]. Numerous studies indicated that mislabeling is a common phenomenon in HRSI classification, typically resulting in reduced classification accuracies. Different ground classes have varying sensitivity levels to spectral reflectance. However, manual labeling of samples may not fully account for the differences in spectral reflectance. Therefore, accurate manual labeling of samples depends on the expertise of practitioners.

Over the past decades, various HRSI classification methods have been proposed to address the associated challenges known as classification with HRSI. Recently, sample augmentation based on deep learning networks for hyperspectral images classification has become attractive in practical applications, such as agricultural monitoring [18], urban planning [19], and environmental assessment [20]. Herein, the related studies were divided into four groups according to the motivation of these methods: sample exploration, sample generation, sample imbalance, and classification neural network with small samples. The details of these methods are presented below.

1) Sample exploration is aimed at searching the potential samples around known samples for sample enhancement. The local pixels of an HRSI typically exhibit homogeneity in spectral and spatial continuities. Accordingly, the potential samples around an initial sample can be explored for sample augmentation, such as superpixel-based sample mining [21], adaptive region expansion-based sample exploration [22], and neighbor-based high-quality sample query [23]. In addition, active learning techniques have been proven to be effective for exploring samples for improving classification performance with HRSI [24]. Majority of the active learning approaches focus on developing different query selection strategies to select unlabeled samples, such as information analysis-based approach [25], uncertainty-based approach [26], and adaptive active learning-based approach [27]. The pseudolabel exploration framework [28], label propagation algorithm [29], and noisy-label detection [30] are also popular in improving classification performance with HRSI via sample enhancement. Methods based on sample exploration can effectively increase the quantity or enhance the quality of samples to improve classification performance. Nevertheless, the explored samples typically aggregate around known samples, which is adverse to capturing the interclass' spectral homogeneity.
2) Sample generation is another effective approach of sample augmentation for improving classification performance with HRSI when the samples are limited [31]. Sample generation aims at producing virtual samples to improve classification accuracies. The simple and commonly used sample generation methods include transformation-based strategies, such as clipping [32], occlusion [33], and rotation [34]. Transformation-based strategies, such as sample augmentation based on random occlusion with CNNs [35] and data augmentation frameworks [36], can effectively increase the quantity and diversity of samples. Sample generation can also be achieved by using a mix-up-based approach, which generates new samples from multiple known samples. Examples of this approach include a CNN-mixture model [37], a soft label mix-up-based CNN [38], and an adversarial MixUp with implicit semantic network [39]. Recently, generative adversarial network (GAN) has attracted significant attention for sample generation, but GAN can generate new samples to simulate the distribution of a known sample set through the competition between the generator and the discriminator [40], [41]. In particular, some derived approaches based on GAN have been proposed for sample generation, including GAN-based distance constraint [42], GAN-based adaptive DropBlock (ADGAN) [43], 3-D-GAN [44], and GAN-based conditional random field [45]. From the literature review, the characteristics of the generated samples depend on the characteristic of the initial sample set. However, the limited sample maybe insufficient to express the entire characteristic of a class. Moreover, some noisy-label or strange samples may be produced by using a sample generation algorithm, which will have a negative influence on the classification performance of a supervised classifier.
3) Besides the quality and quantity of samples, balance of samples plays an important role in HRSI classification [14]. Remote sensing images are one of the typical imbalance data in pattern recognition, and numerous studies have indicated that the interclass imbalance in an image scene requires the corresponding imbalance samples to avoid the bias classification accuracies to the majority of the classes [14]. Various algorithms have been developed to address the class-imbalance problem in the field of land-cover classification. A typical sample balancing algorithms is the oversampling technique [46], which includes random oversampling [47], synthetic minority oversampling technique (SMOTE) [48], and

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distinct variants of SMOTE. SMOTE is a preprocessing technique that effectively relieves the class-imbalance effect. However, oversampling poses challenges in learning the boundaries of minority samples and adjusting the corresponding decisions to minimize class imbalance. Related studies have been proposed, including an adaptive synthetic sampling approach for imbalanced learning [49], borderline-SMOTE [50], and hybrid clustered affinitive borderline-SMOTE [51], to address this problem. Considering the promising performance of deep learning-based approaches even in scenarios of interclass imbalance, such methods are noteworthy for addressing the issue of class imbalance. These approaches include generative adversarial minority oversampling strategies [3 dimensions-hyperspectral generative adversarial minority oversampling (3-D-HyperGAMO)] [52], adaptive drop block-enhanced GANs [43], and deep generative spectral–spatial classifiers [53]. These studies demonstrate that addressing the interclass-imbalance problem in the process of land-cover classification is an effective approach to balance the accuracies of majority and minority classes of an image scene.

4) Considering the promising performance of deep learning neural networks, various deep learning networks have been proposed to achieve satisfactory classification performance when dealing with limited samples [31]. For example, metric-based few-shot learning measures the distance between the samples in a query set and the support set and classifies them by the idea of nearest neighboring samples, such as deep few-shot learning (DFSL) [10], spectral–spatial Siamese network (S3-Net) [54], and multistage relation network with a dual-metric (DM-MRN) [55]. In addition, a cross-domain-based few-shot learning network aims at building a connection between the source domain and the target's domain, which have a strong correspondence, such as cross-domain few-shot learning [56]. Meanwhile, meta-learning [57] and multitask deep learning network (MDL-Net) [58] have been successfully applied for HRSI classification with small samples.

Here, a novel classification approach is proposed for improving classification performance when the initial samples are limited. The proposed approach focuses on two objectives, namely, sample generation and interclass imbalance. The major contributions of this work can be described as follows.

1) The sample generation method based on GAN (SGGAN) was first proposed for generating new samples. In the proposed SGGAN, the spatial–spectral rules are defined to improve the quality of the generated samples. Moreover, certain samples located at the boundary of a class in the generated samples are suggested to be retained to improve the sample's diversity of an intraclass.
2) An iterative sample generation and balance (ISGB) strategy is proposed in order to improve classification performance and balance the user's accuracy for each class via iteration based on the proposed SGGAN. In the proposed ISGB strategy, new samples can be generated for each class, and the quantity of the samples for each class can be balanced through iteration. Moreover, the termination of the iteration is a free-predefined constraint condition for an image.
3) Advanced classification accuracies can be achieved by using the proposed classification approach with HRSI. Experiments carried out on four widely used HRSI datasets effectively demonstrate the feasibility of the proposed classification approach when the initial training samples are limited. When compared with some few-shot learning neural works and GAN-based approaches, the promising performance achieved by the proposed classification approach verified its advantages in HRSI classification.

The remainder of this article is organized as follows. Related work concerning our proposed classification framework is reviewed in Section II. Details of the proposed framework are introduced in Section III. Experiments and results are presented in Section IV, and discussions are provided in Section V. Conclusions are given in Section VI.

II. Related Work

A. Sample Generation

In some prior studies, GAN played an important role in addressing the sample insufficiency in the community of classification with HRSIs [59]. Traditional generative models are limited by nonlinear data, such as restricted Boltzmann machines [60], variational autoencoder [61], and autoregressive model [60]. GAN contains a generator and a discriminator that generates samples by estimating the real sample's distribution. Specifically, the generator generates samples to confuse the discriminator, and the discriminator distinguishes real samples from fake samples. Therefore, the training of discriminator and generator can be regarded as a process of finding Nash equilibrium. For instance, 3-D-GAN uses two 3 dimensions-convolutional neural network (3-D-CNN) networks as the generator and discriminator, fully learns the spectral–spatial information, and fine-tunes the discriminator with the generated samples together with real samples [44]. The GAN-based conditional random field model integrates a semisupervised framework and a probabilistic graphical model for classification [45]. Although various studies indicated that GAN can create new samples for enhancing classification performance, these methods only consider the importance of sample's quantity for HRSI classification, and the negative impact of the imbalance for intraclasses in the sample generation progress is ignored completely [43].

B. Imbalance Correction Techniques

Class imbalance presents a challenge for land-cover classification with HRSIs [52]. The interclass imbalance and intraclass heterogeneity on the Earth's surface are natural and intuitive, and the class imbalance in an image scene is inevitable [62]. Various methods have been proposed for correcting intraimbalance in classification of remote sensing images. The imbalance correction techniques can be

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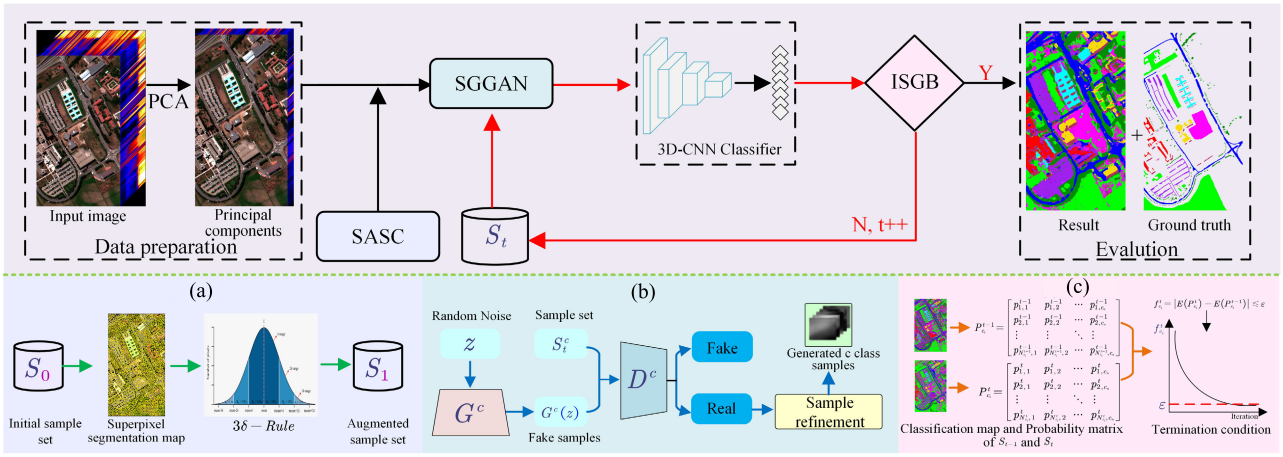


Fig. 1. Flowchart of the proposed approach. (a) SASC: sample augmentation with superpixel's constraint. (b) SGGAN: sample generation based on generative adversarial network. (c) ISGB: iterative sample generation and balance.

divided into two categories: oversampling techniques and deep-learning-based networks. The oversampling techniques focus on balancing the quality between the majority class and the minority class via oversampling techniques. The SMOTE approach, mentioned above, increases the density of minority classes by synthesizing samples of minority-class nearest neighbor samples [48]. Numerous SMOTE-based methods have been widely proposed for addressing the imbalance problem, including borderline-SMOTE [50] and Kmean-SMOTE [63]. However, oversampling-based methods focused solely on the quantity of the samples of each class, and they are not suitable for generating samples in a hyperfeature space. Moreover, the class-imbalance problem has not been effectively addressed using a linear oversampling approach. In addition, deep learning networks can effectively deal with the effect from the imbalanced phenomenon by designing a loss function. Roy et al. [52] proposed a generative adversarial minority oversampling function named 3-D-HyperGAMO to generate samples within the convex hull of the minority class. Notably, 3-D-HyperGAMO is practical for producing synthetic samples and realizing classification in one stage rather than in two separate phases. Nonetheless, the convex hull of a minority class may be far from the true data distribution, which may generate less informative samples [53]. Xi et al. [53] proposed a spatial–spectral distance-based reconstruction loss function to facilitate greater similarity between the known samples and the generated samples. With these methods, it was indicated that designing a suitable loss function for a deep learning neural network can effectively increase the samples of the minority class.

C. Deep Learning Classification Neural Network With Small Samples

Some deep learning classification neural networks have been proposed that are aimed at achieving satisfactory classification performance with HRSI when the samples are limited. For example, the DFSL combines spatial–spectral features with traditional classifiers, such as support vector machine or nearest neighbor classifiers [k nearest neighbors (KNNs)], and uses meta-learning methods to train feature extraction models on accurately labeled samples to achieve HRSI few-shot classification [10]. The S3-Net is constructed with two dual branches that augment training samples by feeding sample pairs into each branch, providing a new approach to achieve classification when the training samples are limited [54]. DM-MRN utilizes a sample recombination strategy to cover the interclass heterogeneity, and an adaptive weighting strategy is designed to fuse the obtained relation scores to improve classification performance [55]. In the existing studies, some deep learning networks have demonstrated a satisfactory classification performance when the training samples are limited. However, sample generation and interclass imbalance are conducted separately in the above studies. Intuitively, sample generation and interclass imbalance are strongly correlated operations, which should be conducted synchronously in a classification task.

III. Method

A. Overview

The overview of the proposed approach for HRSI classification is first presented to provide an entire understanding of the proposed approach. The proposed approach contains three major parts, namely, sample augmentation with superpixel's constraint (SASC), SGGAN, and ISGB strategy (see Fig. 1). The major parts of the proposed approach are elaborated below.

B. SASC

Numerous studies have indicated that the performance of GAN heavily relies on a large amount of samples to tune the generator and discriminator of a GAN [41]. In this section, a sample augmentation approach named SASC is proposed to address the conflict between the limited initial training samples and the samples generated based on GAN. In the proposed SASC, simple linear iterative clustering (SLIC) [64] is adopted to generate superpixels, which are composed of a group of pixels that are spatially continuous and spectrally homogeneous.

Although a superpixel represents a local area with high homogeneity in terms of spectral value, some noise pixels may

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exist within a superpixel. In the proposed SASC, the pixels within the superpixel anchored by a known sample will be marked as potential samples. Thereafter, the marked samples will be refined by a 3δ rule. Some literature has demonstrated that the spectral distribution of the pixels within a local area of remote sensing images is approximately a normal distribution [65]. Here, if the spectral value of a pixel is in the range $[E-3\delta, E+3\delta]$, then it will be marked as a potential sample; otherwise, it will be excluded. E and δ are the mean and standard deviation of the pixels within a superpixel. The proposed SASC provides a strategy to augment the initial sample set to avoid the overfitting of SGGAN in the following steps.

C. SGGAN

An approach named SGGAN was proposed based on the sample augmentation with the proposed SASC. The proposed SGGAN consists of two stages, namely, sample generation and refinement. We define $S^t = \{S^t_{c_1}, S^t_{c_2}, S^t_{c_3}, \ldots, S^t_{c_n}\}$, where S^t denotes the sample set for all the classes (the total number of the classes is n) in an image scene for the tth iteration, and $S^t_{c_i}$ is the sample subset for the class c_i, where $S^t_{c_i} = \{v_1, v_2, v_3, \ldots, v_m\}$, m is the total number of samples for the class c_i, and v_m is the spectral value for the mth sample in the sample subset $S^t_{c_i}$. The two stages of the proposed SGGAN are provided below.

First, in the sample generation stage, samples were generated with GAN class-by-class. For example, GAN_{c_i} generates samples for class c_i, which includes generator G_{c_i} and discriminator D_{c_i}. In particular, the input of generator G_{c_i} is random noise z, which conforms to a Gaussian distribution, and the generated sample set $S^{t'}_{c_i} = G_{c_i}(z)$, where $S^{t'}_{c_i} = \{v'_1, v'_2, v'_3, \ldots, v'_m\}$. The discriminator D_{c_i} receives $S^t_{c_i}$ and $S^{t'}_{c_i}$ and outputs real or fake probability for each generated samples in $S^{t'}_{c_i}$. G_{c_i} attempts to learn the distribution of $S^t_{c_i}$ and generate samples that are subject to the learned distribution. D_{c_i} judges whether the input is real or fake. Therefore, the ultimate goal of the sample generation is to solve the minimax problem, as presented in the following equation:

$$\min_{G_{c_i}} \max_{D_{c_i}} V(D_{c_i}, G_{c_i}) = E_{S^t_{c_i} - p\left(S^t_{c_i}\right)}\left[\log\left(D_{c_i}\left(S^t_{c_i}\right)\right)\right] + E_{z\sim p(z)}\left[\log\left(1-D_{c_i}\left(G_{c_i}(z)\right)\right)\right] \quad (1)$$

where $V(D_{c_i}, G_{c_i})$ represents the similarity between the known sample distribution and the generated sample distribution; and $S^t_{c_i}$ and z are sampled from the known sample distribution $p(S^t_{c_i})$ and Gaussian distribution $p(z)$, respectively. Furthermore, $E(\cdot)$ denotes the expectation operator. When training GAN_{c_i}, an alternating optimization technique is utilized. Specifically, D_{c_i} strived to make $D_{c_i}(S^t_{c_i})$ approach 1 and $D_{c_i}(G_{c_i}(z))$ approach 0 by maximizing $V(D_{c_i}, G_{c_i})$ with G_{c_i} fixed in one iteration. Thereafter, G_{c_i} strives to make $D_{c_i}(G_{c_i}(z))$ approach 1 by minimizing $V(D_{c_i}, G_{c_i})$ with the fixed D_{c_i}. In such an adversarial training, the discriminator D_{c_i} and the generator G_{c_i} promote each other. After several iterations, G_{c_i} can capture the distribution of $S^t_{c_i}$. Moreover, the capability of D_{c_i} in distinguishing the known sample from the generated sample can be enhanced.

Second, to further improve sample diversity, a sample refinement stage is required to refine the generated samples. First, the mean of each class in S^t is calculated and denoted as $\overline{S^t} = \{\overline{S^t_{c_1}}, \overline{S^t_{c_2}}, \overline{S^t_{c_3}}, \ldots, \overline{S^t_{c_n}}\}$. Thereafter, the spectral similarity between the mean of each class and the generated samples is measured by the Euclidean distance. Herein, a sample in $S^{t'}_{c_i}$ is denoted as v'_m. Accordingly, the distance can be provided by $d_{m,i} = \|v'_m - \overline{S^t_{c_i}}\|$. Based on the calculation, if the label of v'_m is consistent with the label of the closest $\overline{S^t_{c_i}}$, v'_m will be retrained as a new sample; otherwise, it will be removed from $S^{t'}_{c_i}$.

Third, the class boundary between different classes should be found and utilized, because selecting some samples at the boundary is beneficial to enhance the distinguishing ability of a classifier and the classification accuracies. Accordingly, each sample v'_m in $S^{t'}_{c_i}$ will be taken as a clustering center, and K neighboring samples in the labeled sample set $S^t_{c_i}$ are chosen. The selected K neighboring samples are the closest samples to the clustering center v'_m in terms of spectral feature space. Herein, a boundary measurement (BM) is borrowed from an existing study [66] to quantitatively measure the extent of a sample belonging to a boundary. The specifics of BM are presented in the following equation:

$$\mathrm{BM}\left(v'_m\right) = \frac{1}{K}\sum_{i=1}^{K}\left(\alpha_i - \frac{2\pi}{K}\right)^2 \quad (2)$$

where K neighboring samples around v'_m can form K angles denoted as $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_K$. The internal samples of the set have a relatively lower BM. Meanwhile, the boundary samples have a higher BM. Thereafter, the samples were sorted in reverse according to the value of BM, and the top-$\mathrm{Num}^t_{c_i}$ samples were selected from $S^{t'}_{c_i}$ as the candidate samples. $\mathrm{Num}^t_{c_i}$ is the total number of generated samples for class ci, and it can be calculated by using the following equation:

$$\mathrm{Num}^t_{c_i} = \frac{E\left(P^t_{c_i}\right)}{\sum_{j=1}^{c_n} E\left(P^t_j\right)} \times w \times h \quad (3)$$

where $w \times h$ denotes the total number of I samples. $\mathrm{Num}^t_{c_i}$ implies that the number of generated samples was determined based on the information entropy ratio of each class.

D. ISGB

SASC is combined with SGGAN into an iterative strategy named ISGB to address the interclass imbalance. The details of the proposed ISGB strategy are shown in Algorithm 1.

Based on the above algorithm, three aspects are worth noting: first, two probability matrix sets are obtained by applying 3-D-CNN based on S^0 and S^1 for the algorithm's initialization. Thereafter, in the $(t-1)$th and tth iterations, each class probability matrix can be calculated by using 3-D-CNN with the corresponding sample set, and the tth probability matrix is shown in the following equation, where $p^t_{N^t_{c_i} c_n}$ is defined as the probability that the $N^t_{c_i}$th sample predicted to be class c_i belongs to class c_n in the tth iteration. The sum of

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Algorithm 1 Algorithm of the Proposed ISGB Strategy

Input: The HRSI image I; the known sample set $S^t = \{S^t_{c_1}, S^t_{c_1}, S^t_{c_1}, \ldots, S^t_{c_n}\}; t \geq 1$; probability sets $P^{t-1} = \{P^{t-1}_{c_1}, P^{t-1}_{c_2}, P^{t-1}_{c_3}, \ldots, P^{t-1}_{c_n}\}$ and $P^t = \{P^t_{c_1}, P^t_{c_2}, P^t_{c_3}, \ldots, P^t_{c_n}\}$ in the $(t-1)-th$ and $t-th$ iteration; Initialize counter $T = 0$; and total class number c_n.

Output: Final classification map M.

Step-1: For i=1, T=0; i≤n; i++; do

1. IF $P^{t-1}_{c_i}$ and $P^t_{c_i}$ dissatisfied termination condition Equation (5):

1.1. $T = T + 1$, calculate the number of generated samples $Num^{t+1}_{c_i}$ in the $(t+1)-th$ iteration of class c_i through Equation (3).

1.2. Construct a new sample set $S^{t+1}_{c_i}$ consisting of $S^t_{c_i}$ and $Num^{t+1}_{c_i}$ samples generated based on SGGAN.

ELSE: $S^{t+1}_{c_i} = S^t_{c_i}$.

End for

Step-2: IF $T \neq 0$:

1. 3D-CNN is applied to produce P^{t+1} based on S^{t+1}.

2. Let $P^{t-1} = P^t$, $P^t = P^{t+1}$ and $t = t + 1$.

3. Go to Step-1.

ELSE: The algorithm is terminated, and the final classification map M is obtained based on P^t.

each row of the probability matrix is equal to one

$$P^t_{c_i} = \begin{bmatrix} p^t_{1,1} & p^t_{1,2} & \cdots & p^t_{1,c_n} \\ p^t_{2,1} & p^t_{2,2} & \cdots & p^t_{2,c_n} \\ \vdots & \vdots & \ddots & \vdots \\ p^t_{N^t_{c_i},1} & p^t_{N^t_{c_i},2} & \cdots & p^t_{N^t_{c_i},c_n} \end{bmatrix}. \tag{4}$$

Second, the termination condition of the iteration can be determined by using the following equation:

$$\left| E\left(P^t_{c_i}\right) - E\left(P^{t-1}_{c_i}\right) \right| \leq \varepsilon \tag{5}$$

where ε is a small constant and $E(\cdot)$ is the average information entropy function that indicates the average level of information inherent from class c_i. Equation (5) implies that the sample generation for class c_i will be terminated when the difference between two probability matrixes from two adjacent iterations in terms of the average information entropy is approximate. $E(\cdot)$ can be calculated by using the following equation:

$$E\left(P^t_{c_i}\right) = -\frac{\sum_{i=1}^{N^t_{c_i}} \sum_{j=1}^{c_n} p^t_{i,j} \log p^t_{i,j}}{N^t_{c_i}} \tag{6}$$

where $N^t_{c_i}$ and $P^t_{c_i}$ are the number of samples and probability matrix predicted for class c_i at the tth iteration.

From the detailed introduction of the three major parts of the proposed approach for classification with HRSI, the proposed approach combines the new approaches related to sample augmentation, generation, and balancing strategy to improve the classification performance with HRSI when the initial training samples are limited. In the above presentation, to solve the overfitting of generating sample based on very small training samples, a simple yet effective SASC was proposed to enhance the quantity of the initial sample set. The proposed SGGAN module aims at generating samples based on GAN and improved sample diversity. Its advantage lies in that it can generate sufficient samples for training a network. However, when the initial samples are insufficient, the distribution samples based on SGGAN may gradually deviate from the real distribution of a class in the iterative progress. The proposed ISGB concentrates on coupling with SGGAN in iterative progress to dynamically balance the samples between different classes, and experiments well verified the proposed approach can improve classification performance and balance users' accuracy. However, the proposed ISGB is an iterative algorithm, which requires high time cost when processing a large dataset in practical applications.

IV. Experiment

Two experiments are here organized to verify the feasibility and advantages of the proposed classification approach with HRSI when the training samples are limited. The experiments include the comparison with some state-of-the-art approaches and discussion about the relationship between the classification accuracies and the parameters of the proposed approach. The experimental details are presented below.

A. Datasets Description

Four widely used HRSI datasets, including Pavia University (Dataset-1), Salinas (Dataset-2), WHU-Hi-HanChuan (Dataset-3), and Houston University (Dataset-4), are adopted to evaluate the feasibility and advantages of the proposed method. The detailed description of each dataset is summarized in Table I, and the false-color image and the corresponding ground reference map are illustrated in Fig. 2. These datasets are acquired by using different sensors, including reflective optics system imaging spectrometer (ROSIS) sensor, airborne visible/infrared imaging spectrometer (AVIRIS) sensor, and Headwall Nano-Hyperspec imaging sensor, with a spatial resolution coverage ranging from 0.1 to 3.7 m/pixel. In addition, the datasets covered different ground scenes, such as the university area (Dataset-1 and Dataset-4) and crop area (Dataset-2 and Dataset-3), therefore achieving satisfactory classification maps using a limited number of samples for the datasets.

B. Experimental Settings

Three representative methods based on GAN for HRSI classification, 3-D-GAN [44], 3-D-HyperGAMO [52], and ADGAN [43], are compared with the proposed approach to verify the feasibility and advantages of the proposed framework. In addition, some few-shot learning methods, including 3-D-CNN [67], DFSL [10], S3-Net [54], MDL-Net [58], and multistage deep relation network (MD-RN) [55], are selected for comparison. Each approach was briefly described below.

1) *3-D-GAN [44]:* This is a spectral–spatial classification model. The GAN is used with small real training samples to fine-tune the discriminative CNN, which improves the classification performance.
2) *3-D-HyperGAMO [52]:* This model uses the generative adversarial minority oversampling technique, which

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TABLE I
DETAILED DESCRIPTION FOR EACH DATASET

Dataset	Location	Platform and Sensor	Size (pixel)	Resolution (m/pixel)	Total classes
Pavia University (Dataset-1)	Pavia University, Italy	Aerial planes and ROSIS sensor	$610 \times 340 \times 103$	0.32	9
Salinas (Dataset-2)	California, USA	AVIRIS sensor	$512 \times 217 \times 224$	3.70	16
WHU-Hi-HanChuan (Dataset-3)	Hubei, China	Headwall Nano-Hyperspec imaging sensor	$1217 \times 303 \times 274$	0.10	16
Houston University (Dataset-4)	Houston University, USA	Compact airborne spectrographic imager	$601 \times 2384 \times 48$	2.50	20

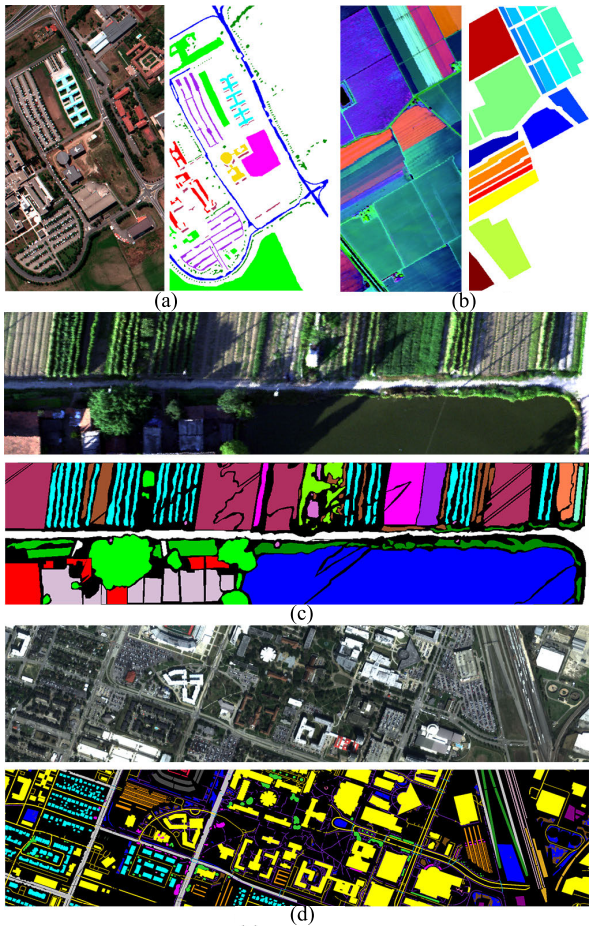


Fig. 2. Dataset used for experiments with three-band false-color images. (a) Dataset-1. (b) Dataset-2. (c) Dataset-3. (d) Dataset-4.

automatically generates high-quality samples for minority classes at the training stage using the existing samples of that class.

3) *ADGAN [43]:* This network contains a single output in a discriminator, which proves effective in dealing with minority-class samples. Furthermore, ADGAN generates masks with adaptive shapes, which can boost the classification performance.
4) *3-D-CNN [67]:* A 3-D deep learning approach enables joint spectral and spatial information to improve the classification performance.
5) *DFSL [10]:* A DFSL method—the key idea is that the designed network learns a metric space from the training dataset where samples from the same class are close, and those from different classes are far.
6) *S3-Net [54]:* This network has two branches to augment samples by feeding sample pairs into each branch, enhancing the model separability, where negative samples are randomly selected to avoid redundancy.
7) *MDL-Net [58]:* This network is developed to identify unknown classes. The distance between classes can be maximized to achieve classification by using extreme value theory to reconstruct the loss. This approach is particularly advantageous in scenarios with small samples.
8) *MD-RN [55]:* This network concentrates on the underutilization of scarce labeled samples. A sample recombination strategy is designed to increase the effectiveness and robustness of the classification model.

Ten pixels are randomly selected from the ground reference map for the initial samples to guarantee comparative fairness. The default values of the parameters of each approach are used, as per the related literature. For instance, the classification map is obtained by running 3-D-CNN for 100 epochs. In our proposed approach, a patch of size $9 \times 9 \times 10$ is extracted from Dataset-1 to Dataset-4. The spatial dimension of the random noise vector is fixed at 100, $K = 5$, and $\varepsilon = 0.01$. All the experiments are implemented in Python 3.7 and PyTorch 1.13 with a Geoforce NVIDIA 1080Ti.

C. Evaluation Metrics

Four widely used indicators, including overall accuracy (OA) for all samples, average accuracy (AA) for the classes, Kappa coefficient (Ka), and $F1$ score, are adopted in the following experiments for a quantitative comparison. In addition, the standard deviation of the user's accuracy (SDUA) is adopted to evaluate the accuracy-balanced ability for each class. Details of each metric are summarized in Table II, where N is the total number of classes in an image scene, UA_i is the user accuracy for the ith class, and $\overline{\mathrm{UA}}$ is the mean value of user's accuracy for all the classes.

D. Experimental Results

The proposed approach is compared with eight deep learning methods based on the above datasets and experimental setting. The classification accuracies based on different methods are summarized in Tables III–VI. The quantitative comparisons indicated that the proposed approach not only achieves superior accuracies but also exhibits robust superiority in quantitative accuracies across different datasets. For example, the proposed approach obtained OA = 93.58%, AA = 92.87%, Ka = 0.916, and $F1$ score = 93.73%, which are the best accuracies when comparing with the eight state-of-the-art methods for Dataset-1. Moreover, the proposed approach obtained SDUA = 6.98, demonstrating competitiveness with the best SDUA = 6.70 obtained by MDL-Net [58]. A similar conclusion can be obtained when

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TABLE II

BRIEF SUMMARY OF THE APPLIED EVALUATION METRICS

Evaluation Indicators	Formula	Explanation
Overall Accuracy (OA)	$OA = \frac{TP+TN}{TP+TN+FP+FN}$	OA is the correctly detected pixels between the total pixels of the ground map.
Average Accuracy (AA)	AA = $(\frac{TP}{TP+FP} + \frac{TN}{TN+FN})/2$	AA is the mean of correctly detected changed and correctly unchanged ratios.
Kappa Coefficient (Ka)	$P_0 = \frac{TP+TN}{TP+TN+FP+FN}$ $P_e = \frac{(TP+FP)\times(TP+FN)+(FN+TN)\times(FP+TN)}{(TP+TN+FP+FN)^2}$ $Ka = \frac{P_0-P_e}{1-P_e}$	Ka reflects the reliability of the detection map by measuring inter-rater reliability for changed and unchanged classes.
F1-score (F1-score)	$F1 = 2 \times \frac{Pr \times Re}{Pr+Re}$	F1-score is the harmonic mean of precision and recall.
Standard Deviation User's Accuracy (SDUA)	$SDUA = \sqrt{\frac{1}{T}\sum_{i=1}^{i=T}(UA_i - \overline{UA})^2}$	SDUA is the standard deviation of user's accuracy.

TABLE III

QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH FOR DATASET-1 (# DENOTES THE TOTAL ITERATIONS OF THE PROPOSED APPROACH). THE BEST RESULTS ARE GIVEN IN BOLD

Method	OA (%)	AA (%)	Ka	F1-score (%)	SDUA
3D-GAN [44]	82.24	84.44	0.756	74.43	11.92
3D-HyperGAMO [52]	90.86	87.59	0.878	85.64	9.08
ADGAN [43]	88.96	82.64	0.856	83.78	15.41
3D-CNN [67]	80.64	77.70	0.750	79.55	11.67
DFSL [10]	85.41	88.34	0.810	85.67	13.12
S3-Net [54]	89.87	87.03	0.866	88.17	9.57
MDL-Net [58]	90.95	87.33	0.881	89.10	**6.70**
MD-RN [55]	84.98	77.27	0.803	78.22	14.80
Proposed (#11)	**93.58**	**92.87**	**0.916**	**93.73**	6.98

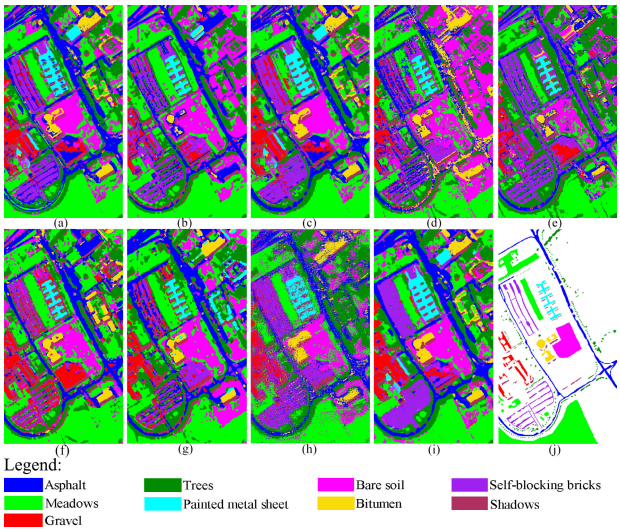


Fig. 3. Obtained classification maps using different traditional methods for Dataset-1 along with the ground truth. (a) 3-D-GAN [44]. (b) 3-D-HyperGAMO [52]. (c) ADGAN [43]. (d) 3-D-CNN [67]. (e) DFSL [10]. (f) S3-Net [54]. (g) MDL-Net [58]. (h) MD-RN [55]. (i) Our proposed approach. (j) Ground truth.

observing the quantitative comparison in Tables IV–VI for Dataset-2, Dataset-3, and Dataset-4, respectively.

When we observed the visual performance comparisons (see Figs. 3–6), the proposed approach has three advantages in improving the visual performance of classification: 1) the classification maps achieved by the proposed approach have less salt-and-pepper noise; 2) the boundary among different classes can be well preserved by the proposed approach; and 3) the shape and size of the ground targets in the classification maps achieved by our proposed approach can be well described, such as the asphalt and the self-blocking bricks in the classification map based on Dataset-1. The proposed approach

TABLE IV

QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH FOR DATASET-2 (# DENOTES THE TOTAL ITERATIONS OF THE PROPOSED APPROACH). THE BEST RESULTS ARE GIVEN IN BOLD

Method	OA (%)	AA (%)	Ka	F1-score (%)	SDUA
3D-GAN [44]	89.55	94.70	0.883	92.61	10.55
3D-HyperGAMO [52]	91.00	93.33	0.900	93.65	8.16
ADGAN [43]	90.95	94.15	0.899	92.60	12.97
3D-CNN [67]	86.92	93.83	0.855	93.49	12.86
DFSL [10]	88.81	92.73	0.875	90.60	8.79
S3-Net [54]	93.85	95.38	0.931	**96.28**	**5.27**
MDL-Net [58]	92.82	94.84	0.920	95.05	7.72
MD-RN [55]	88.46	88.01	0.871	88.78	10.92
Proposed (#15)	**94.19**	**95.57**	**0.935**	96.18	5.76

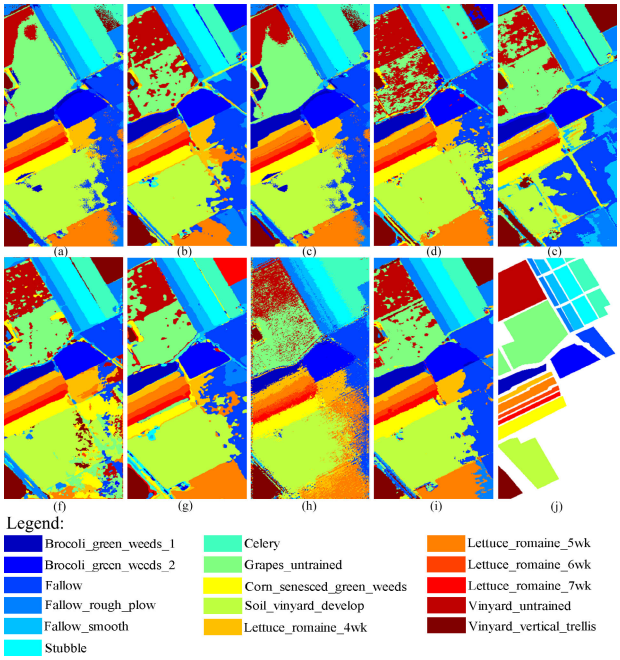


Fig. 4. Obtained classification maps using different traditional methods for Dataset-2 along with the ground truth. (a) 3-D-GAN [44]. (b) 3-D-HyperGAMO [52]. (c) ADGAN [43]. (d) 3-D-CNN [67]. (e) DFSL [10]. (f) S3-Net [54]. (g) MDL-Net [58]. (h) MD-RN [55]. (i) Our proposed approach. (j) Ground truth.

generated samples by considering the spectral homogeneity of interclasses and explored the samples from the boundary among different classes.

V. DISCUSSION

The relationship between the parameters of the proposed approach and the classification accuracy is discussed now to

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TABLE V

QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH FOR DATASET-3 (# DENOTES THE TOTAL ITERATIONS OF THE PROPOSED APPROACH). THE BEST RESULTS ARE GIVEN IN BOLD

Method	OA (%)	AA (%)	Ka	F1-score (%)	SDUA
3D-GAN [44]	75.88	66.48	0.718	59.19	21.6
3D-HyperGAMO [52]	82.01	67.04	0.790	66.88	21.19
ADGAN [43]	79.11	69.67	0.755	63.01	21.51
3D-CNN [67]	73.29	59.88	0.693	61.84	20.45
DFSL [10]	80.52	67.20	0.774	70.07	21.17
S3-Net [54]	83.53	72.67	0.809	75.61	19.82
MDL-Net [58]	84.76	73.62	0.822	76.06	19.87
MD-RN [55]	80.01	68.39	0.769	70.68	21.96
Proposed (#14)	**85.83**	**75.01**	**0.834**	**77.41**	**19.23**

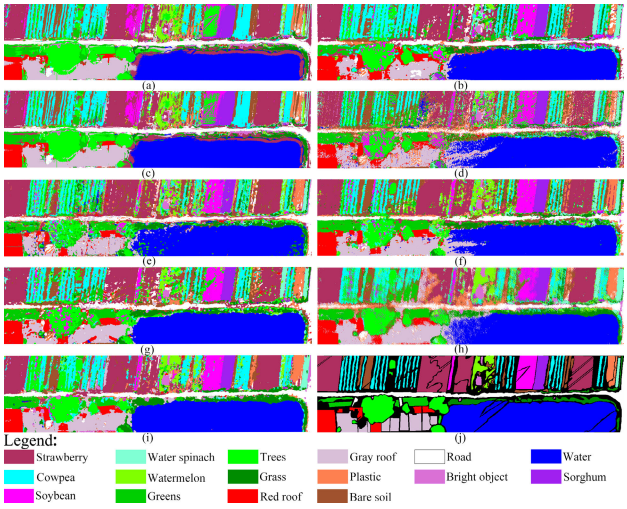


Fig. 5. Obtained classification maps using different traditional methods for Dataset-3 along with the ground truth. (a) 3-D-GAN [44]. (b) 3-D-HyperGAMO [52]. (c) ADGAN [43]. (d) 3-D-CNN [67]. (e) DFSL [10]. (f) S3-Net [54]. (g) MDL-Net [58]. (h) MD-RN [55]. (i) Our proposed approach. (j) Ground truth.

TABLE VI

QUANTITATIVE COMPARISON BETWEEN THE SELECTED METHODS AND THE PROPOSED APPROACH FOR DATASET-4 (# DENOTES THE TOTAL ITERATIONS OF THE PROPOSED APPROACH). THE BEST RESULTS ARE GIVEN IN BOLD

Method	OA (%)	AA (%)	Ka	F1-score (%)	SDUA
3D-GAN [44]	45.08	37.79	0.375	42.59	27.31
3D-HyperGAMO [52]	48.83	45.54	0.422	**49.73**	23.04
ADGAN [43]	48.63	40.84	0.414	45.37	22.99
3D-CNN [67]	41.74	**46.45**	0.293	27.17	26.32
DFSL [10]	50.09	45.06	0.330	25.92	24.72
S3-Net [54]	50.21	42.23	0.275	23.38	23.72
MDL-Net [58]	52.61	48.29	0.366	29.83	25.25
MD-RN [55]	48.76	36.35	0.417	41.28	27.05
Proposed (#12)	**53.75**	43.98	**0.492**	47.21	**22.79**

facilitate the widespread adoption of the proposed approach in practical applications. The discussion below focuses on four items.

1) The size of the superpixels was included in the proposed framework, and a change in the size may affect the classification performance. Let us define a segmentation coefficient φ for SLIC, where $\varphi = (w \times h/\text{seg})$; w and h are the width and height of an image scene, respectively; and seg denotes the total segments obtained by the SLIC.

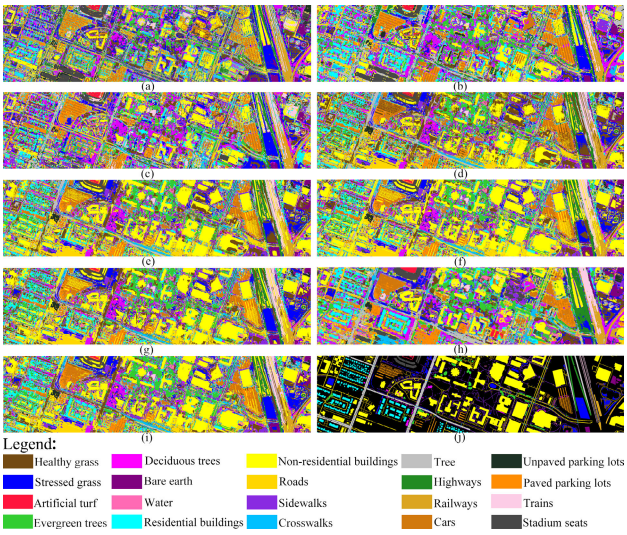


Fig. 6. Obtained classification maps using different traditional methods for Dataset-4 along with the ground truth. (a) 3-D-GAN [44]. (b) 3-D-HyperGAMO [52]. (c) ADGAN [43]. (d) 3-D-CNN [67]. (e) DFSL [10]. (f) S3-Net [54]. (g) MDL-Net [58]. (h) MD-RN [55]. (i) Our proposed approach. (j) Ground truth.

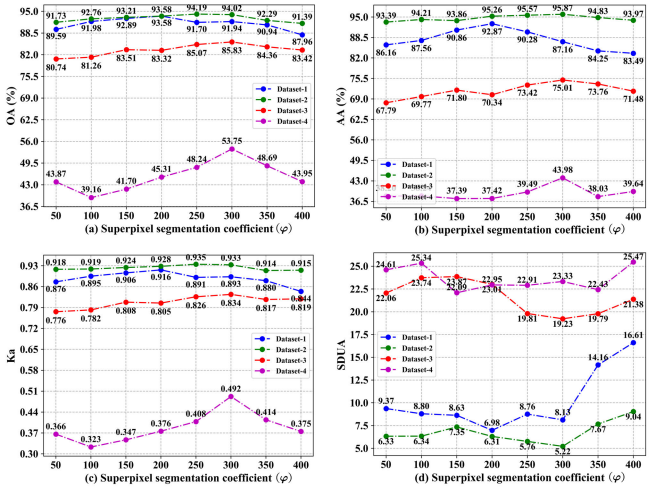


Fig. 7. Relationship between the classification accuracies and the superpixel segmentation coefficient of the proposed approach. (a) Relationship between classification overall accuracy and the superpixel segmentation coefficient for Dataset-1 to Dataset-4. (b) Relationship between classification average accuracy and the superpixel segmentation coefficient for Dataset-1 to Dataset-4. (c) Relationship between classification Kappa and the superpixel segmentation coefficient for Dataset-1 to Dataset-4. (d) Relationship between classification SDUA and the superpixel segmentation coefficient for Dataset-1 to Dataset-4.

We now investigate the relationship between the segmentation coefficient and the class classification accuracy. It is observed that different datasets exhibit varying fluctuations with the increment of the superpixel segmentation coefficient in terms of the diverse evaluation metrics (see Fig. 7). For example, OA increased from 89.59% to 93.58% when the segmentation coefficient increased from $\varphi = 50$ to $\varphi = 200$ for Dataset-1. In particular, the OA increased from 43.87% to 53.75% when the segmentation coefficient increased from $\varphi = 50$ to $\varphi = 300$ for Dataset-4. OA gradually decreased with the continued increase of φ, because an overlarge segmentation means fewer segments of an input image

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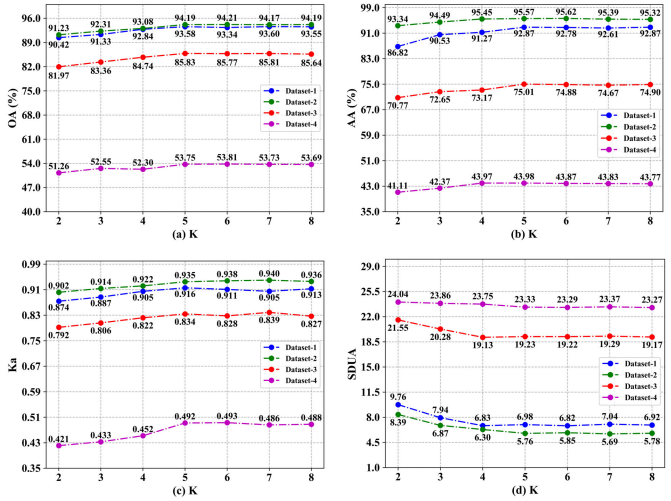


Fig. 8. Relationship between the classification accuracies and the parameter K of the proposed approach. (a) Relationship between classification overall accuracy and the parameter K of the proposed approach for Dataset-1 to Dataset-4. (b) Relationship between classification average accuracy and the parameter K of the proposed approach for Dataset-1 to Dataset-4. (c) Relationship between classification Kappa and the parameter K of the proposed approach for Dataset-1 to Dataset-4. (d) Relationship between classification SDUA and the parameter K of the proposed approach for Dataset-1 to Dataset-4.

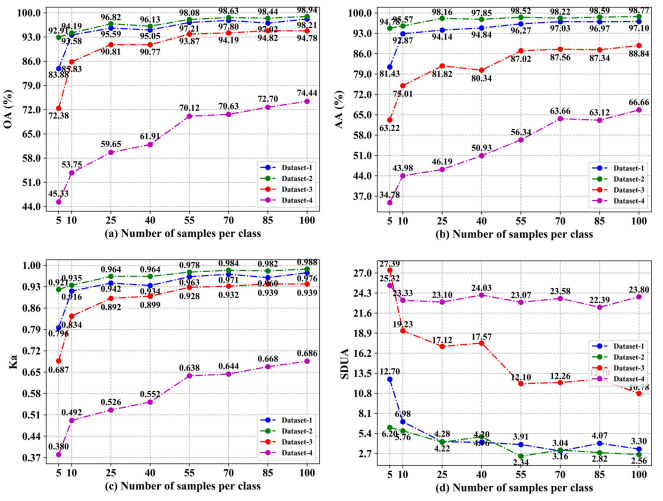


Fig. 9. Relationship between the classification accuracies and the number of initial training samples of the proposed approach. (a) Relationship between classification overall accuracy and the number of initial training samples of the proposed approach for Dataset-1 to Dataset-4. (b) Relationship between classification average accuracy and the number of initial training samples of the proposed approach for Dataset-1 to Dataset-4. (c) Relationship between classification Kappa and the number of initial training samples of the proposed approach for Dataset-1 to Dataset-4. (d) Relationship between classification SDUA and the number of initial training samples of the proposed approach for Dataset-1 to Dataset-4.

scene, and an overlarge segment may reduce the spectral homogeneity of a local area. Furthermore, selecting potential samples within a superpixel may result in some mislabeled samples, which will reduce the classification accuracy. Therefore, an optical segmentation coefficient should be obtained by trial-and-error experiments when applying our approach in practical applications.

2) The relationship between the parameter K of the proposed approach and the classification accuracies was investigated, as shown in Fig. 8. The investigation indicated that the classification accuracies improved slightly with an increase in the value of the parameter K. For example, OA improved from 90.42% to 93.58% when K increased from 2 to 5 for Dataset-1, and AA improved from 93.34% to 95.57% when K increased from 2 to 5 for Dataset-2. A larger K results in more samples being explored around a known sample, but that also means more time cost. Therefore, a trade-off between the degree of improvement and the setting of K should be considered when applying the proposed approach in practical applications. Here, we set $K = 5$ for all datasets in the following experiments.

3) The total number of initial samples for each class plays an important role in the performance of the proposed approach. The relationship between the number of samples of per class and the classification accuracy was investigated (see Fig. 9). The results of the investigation indicated that the performance of the approach was improved with the increase in the initial number of samples per class. However, the improvement is small when the initial samples are sufficient to guarantee that the proposed approach acquires satisfactory classification accuracies. For example, the OA classification accuracy increased from 83.88% to 93.58% when the number of samples per class increased from 5 to 10. Thereafter, the OA ranged from 93.58% to 98.21% when the number of samples per class increased from 10 to 100. In a similar way as for numerous classification methods, increasing the initial training samples per class improves the classification performance of the proposed approach intuitively, but it has an advantage when the initial samples are less and equal to ten samples per class. Moreover, the relationship between SDUA and the number of initial samples per class indicates that the balancing ability of the proposed approach can be improved with an increase in the number of initial samples per class. The various evaluation indicators exhibit diverse sensitivity levels to the changes in the number of initial samples per class in terms of different datasets. Therefore, a trade-off between the classification accuracies and the labeling of initial samples is required when applying the proposed approach in practical applications.

TABLE VII
ABLATION EXPERIMENT ON DIFFERENT DATASETS

Dataset	Module Ablation	OA (%)	AA (%)	Ka	SDUA
Dataset-1	Basic	69.60	59.23	0.591	25.72
	Basic+SASC	85.43	78.63	0.812	15.71
	Basic+SASC+(SGGAN+ISGB)	**93.58**	**92.87**	**0.916**	**6.98**
Dataset-2	Basic	74.14	82.73	0.713	22.39
	Basic+SASC	87.71	90.23	0.862	11.62
	Basic+SASC+(SGGAN+ISGB)	**94.19**	**95.57**	**0.935**	**5.76**
Dataset-3	Basic	51.31	40.59	0.439	27.17
	Basic+SASC	77.81	62.92	0.742	24.56
	Basic+SASC+(SGGAN+ISGB)	**85.83**	**75.01**	**0.834**	**19.23**
Dataset-4	Basic	33.32	29.40	0.258	27.79
	Basic+SASC	45.47	38.67	0.376	25.73
	Basic+SASC+(SGGAN+ISGB)	**53.75**	**43.98**	**0.492**	**22.79**

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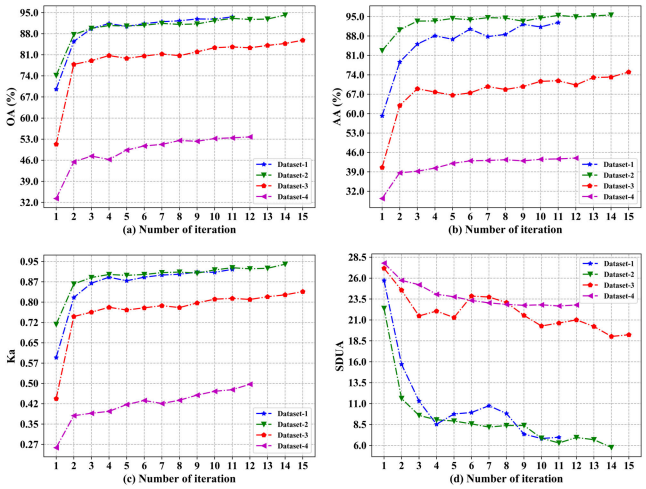


Fig. 10. Relationship between the classification accuracies and the iterations of the proposed approach. (a) Relationship between classification overall accuracy and the iterations of the proposed approach for Dataset-1 to Dataset-4. (b) Relationship between classification average accuracy and the iterations of the proposed approach for Dataset-1 to Dataset-4. (c) Relationship between classification Kappa and the iterations of the proposed approach for Dataset-1 to Dataset-4. (d) Relationship between classification SDUA and the iterations of the proposed approach for Dataset-1 to Dataset-4.

4) The ablation experiment is discussed now to verify the feasibility of the proposed modules. We verify the importance of each module by adding each module to the basic framework one by one, including the SASC module and the SGGAN + ISGB module. The addition of an SASC module to the basic framework brings significant performance improvements, as detailed in Table VII. For instance, in comparison with the basic framework, the OA metrics increased by 15.83%, 13.57%, 26.50%, and 12.15% on the four datasets, reaching 85.43%, 87.71%, 77.81%, and 45.41%, respectively. The performance improvement is attributed to the SASC module, which expands training samples and introduces less noise. In addition, coupling the proposed SASC module and the SGGAN + ISGB module with the basic framework achieved the best classification accuracies. This combination outperformed individual use of the SASC module, because the SASC module and SGGAN + ISGB can remove noise samples while effectively generating samples at class boundaries, resulting in significant performance improvements.
5) The relationship between the iteration of the proposed framework and the classification accuracy is now discussed. The iteration of the proposed approach was determined by a prediction condition, with the parameter in the condition set to a small value, fixed at 0.01 in all the experiments. The relationship between the iteration of the proposed approach and the classification approach was discussed here, based on the clarification. In Fig. 10, the accuracies increased in the majority of cases with the increment of the iteration times from one to four. Thereafter, the accuracies slightly changed and tended to be at a horizontal level after the increment of iterations was larger than four. By contrast, SDUA was decreased with the iteration from one to four, and then, it also tended to a horizontal level when the iteration was larger than four. The observation we can make is that the proposed approach terminated at various iterations for the different datasets. The observation demonstrated the following: a) the augmented samples using the proposed approach are effective for improving classification accuracies with HRSIs and b) the predefined iteration termination conditions are effective in dynamically adjusting the termination of the iteration of the proposed approach and balancing the user's accuracy for the different datasets.

VI. Conclusion

In this work, the proposed method can effectively generate samples and obtain satisfactory HRSI classification performance when initial samples are limited. One advantage of the proposed approach is that it requires small initial samples for real dataset classification, such as it only needs 10 pixels/class for Dataset-1 to Dataset-4, but obtains OA = 93.58%, OA = 94.19%, OA = 85.83%, and OA = 53.75%, respectively, which are better than that of some state-of-arts methods. In addition, the proposed approach has an advantage for balancing the user's accuracies, such as the SDUA of the proposed method. It is significantly reduced by 18.74, 16.63, 7.94, and 5.0 for the four datasets compared with that of the initial classification map.

Although the proposed classification approach demonstrates advantages in improving land-cover classification performance for HRSI when the initial training samples are limited, the method exhibited certain limitations when applied in practical applications. For example, it involves four parameters, the optimized determination of which can be time-consuming. In addition, large-area datasets are required to further investigate the usability of the proposed approach. Therefore, in future work, we will work on developing an estimation approach for parameter estimation to reduce the dependency on manual parameter tuning. In addition, the plan is to collect large-area datasets to further verify the usability of the approach.

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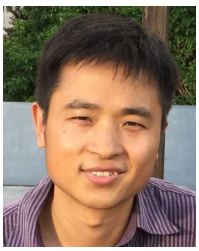
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